ACM CHIIR 2027 · Short Paper

Show the future honestly —

and let people choose how to love what they love



OshiFit: Possible-Selves Avatars for Uncertainty-Aware Behavior Change in Fan Activities

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Oshi-katsu: a booming fan culture

Media Interaction Laboratory, Osaka Institute of Technology

¥3.8T

estimated market for fan activities in Japan (NRI, 2025)

1 in 3

people in Japan engage in oshi-katsu

Well-being

fan activities can foster purpose and mental health

…but also “fandom fatigue”

- Overspending money and time
- Comparing yourself with other fans on social media
- Burnout from a sense of obligation



Preliminary survey (N = 215)

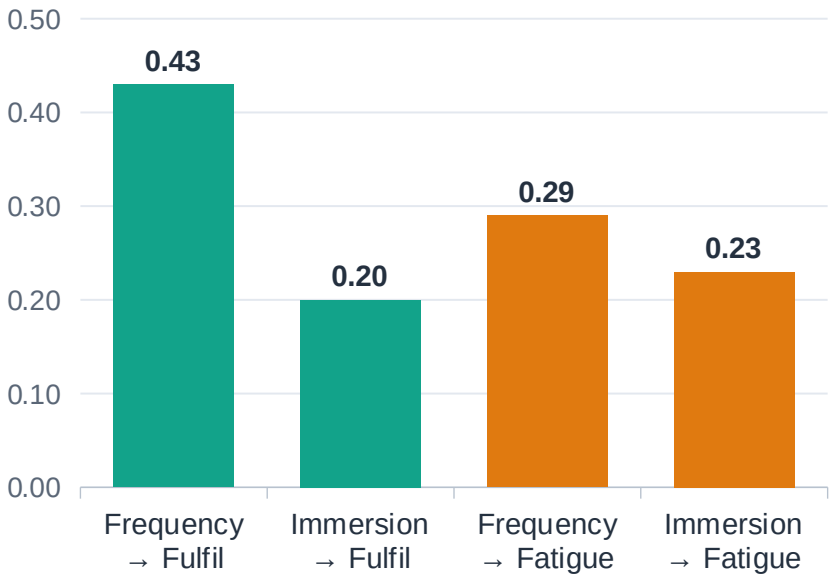
~80%
oshi = reason to live

~40%
worry about money

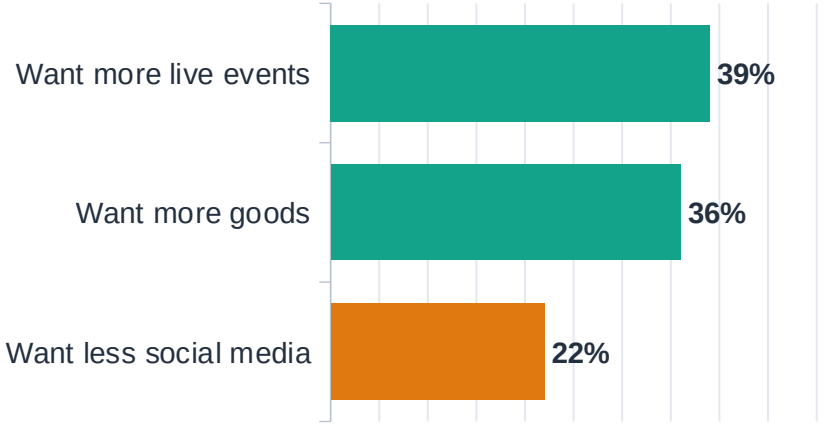
~30%
lose sleep

27%
are high in BOTH fulfilment and fatigue
The same activities drive joy and exhaustion

Path coefficients (personality |β| ≤ .21)



More people want MORE than less



The challenge: noisy forecasts

Our forecast model explains $R^2 = .27$ of fulfilment and only $R^2 = .11$ of fatigue (±0.92 on a 5-point scale)

Showing one confident “future you” would overclaim — and could cause needless anxiety

Personalize

A Bayesian twin starts from the survey and learns from each person's daily logs

Show uncertainty

A possible-selves avatar renders the forecast range as overlapping selves

Rebalance, not restrict

Proposals re-allocate activities and never cut what people want more of

Goal: support behavior change without overclaiming — and without taking the joy away

OshiFit at a glance

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TIPI-J
once, 1 min

Daily log
~30 s a day

Wishes
more / less of what

①

Bayesian Twin

Personal model, updated nightly

②

Rebalance Engine

Re-allocate, not restrict: 3 Pareto options

③④ Future Mirror


As is


Proposal

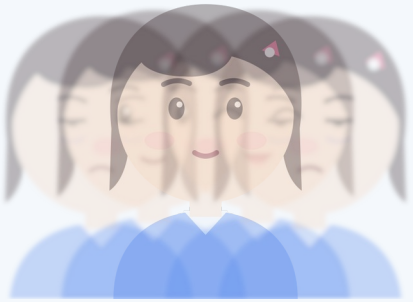
Possible-selves avatar
range shown as overlap

Fatigue-facet cues
sleep, money, duty, compare

Choose → try a week → check-in → log → nightly update → forecast check

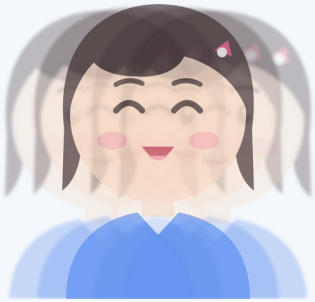
Safety layer: when logs signal strong distress, forecasts pause and help resources are shown

Possible-selves avatar



Day 0

a blurry future



Day 7

selves converge



Day 14

a sharper you

How

Draw 20 futures from the posterior, bin them into 3 states, overlay 5 selves. Opacity = likelihood

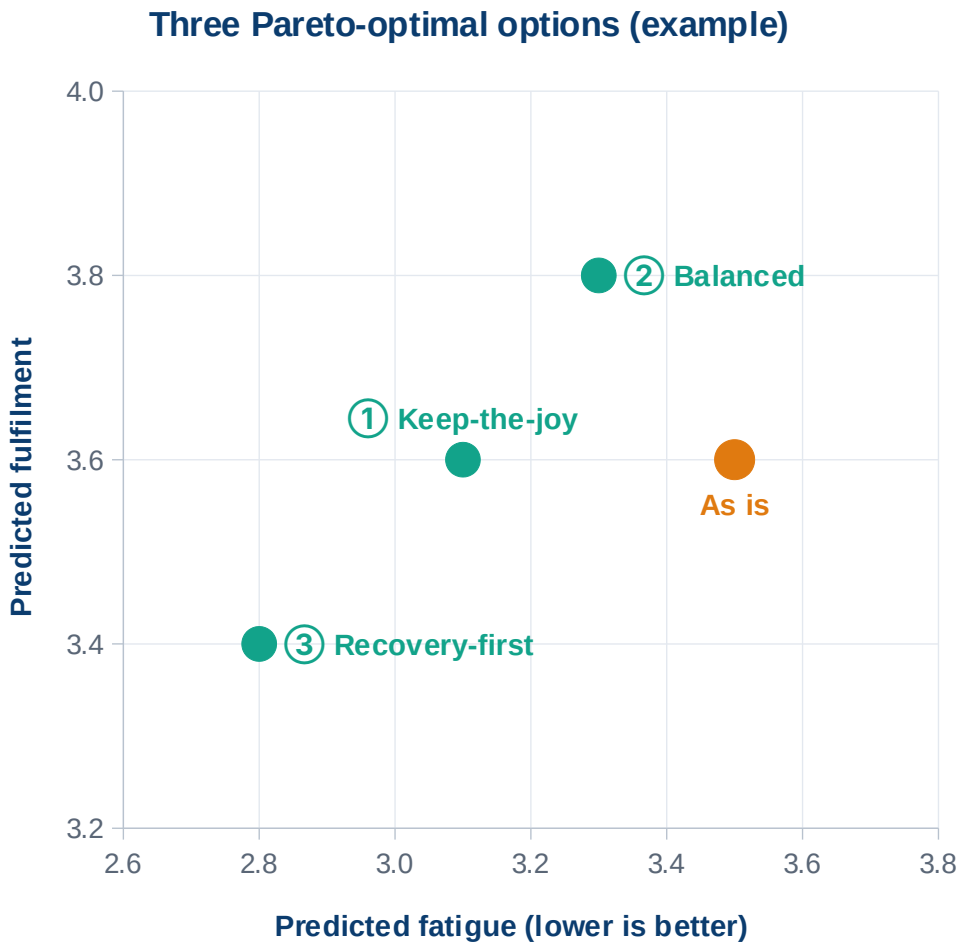
Why

Uncertainty becomes visible, and “future-self resolution” rewards continued logging

Grounded in

Proteus effect, future-self renderings (Hershfield et al., 2011), quantile dotplots (Kay et al., 2016)

Rebalance, not restrict



How options are made

- 98 candidates: change up to 2 of 7 activities by one step
- Drop options that cut what the user wants more of
- Keep the Pareto front; show 3 options

The user chooses

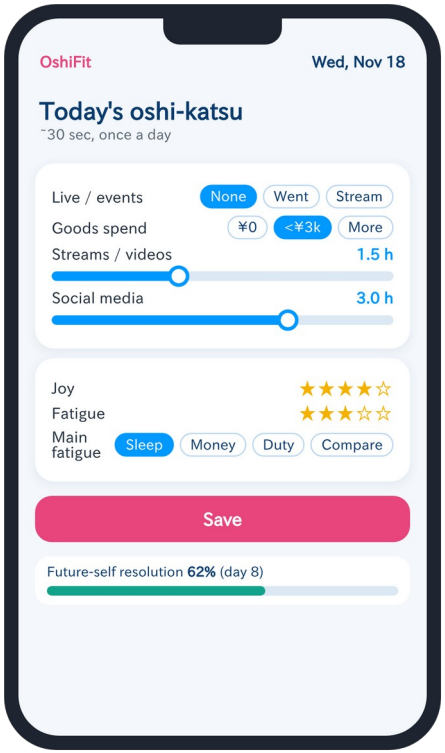
…or skips the week. Autonomy is part of the design

※ 予測値は例示 (example values)

A week with OshiFit

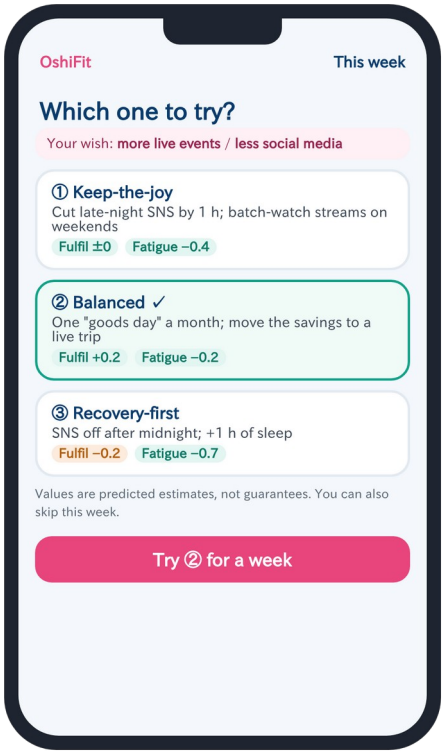
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Every day



Log in ~30 s

Weekly



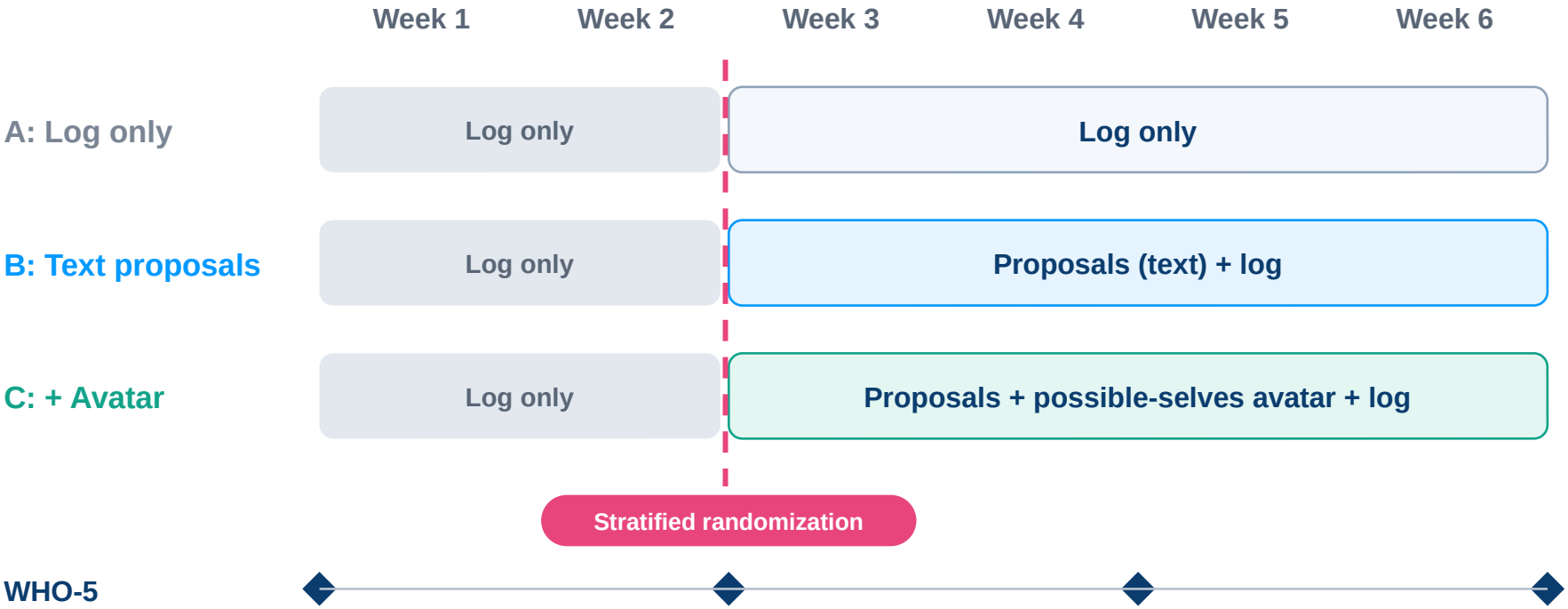
Pick a proposal

Weekly (Group C)



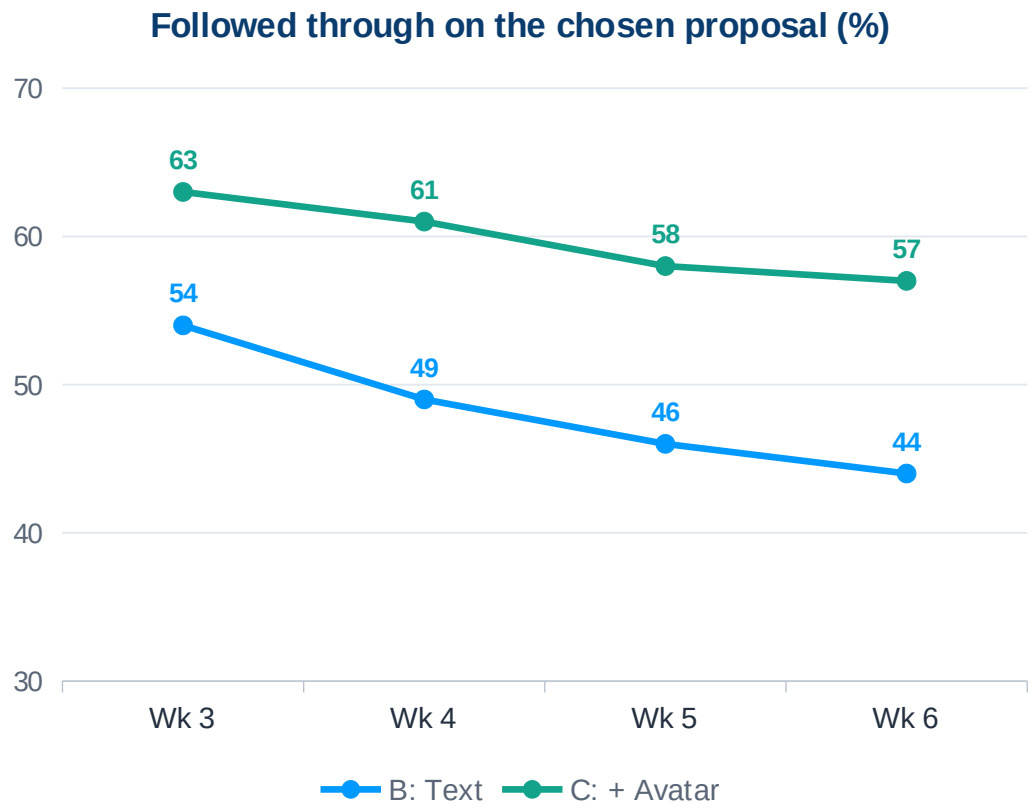
Compare futures

Study: 6-week, three-arm RCT



Measure	Analysis
RQ3 (primary)	WHO-5 at week 6, ANCOVA adjusting for week 2; C vs B and B vs A with Holm correction
RQ2 · RQ1	Weekly follow-through (logistic mixed model) · forecast error at week 6

Results: avatars boosted follow-through



N = 197 completed (after the paper was submitted). Same proposals in B and C.

OR = 1.71

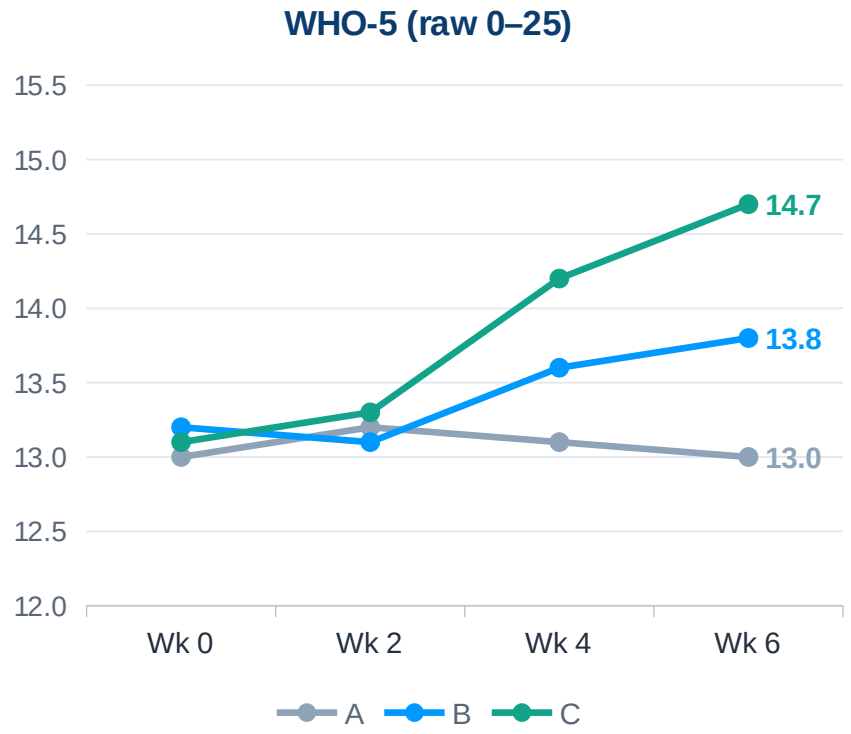
95% CI 1.18–2.48, p = .005

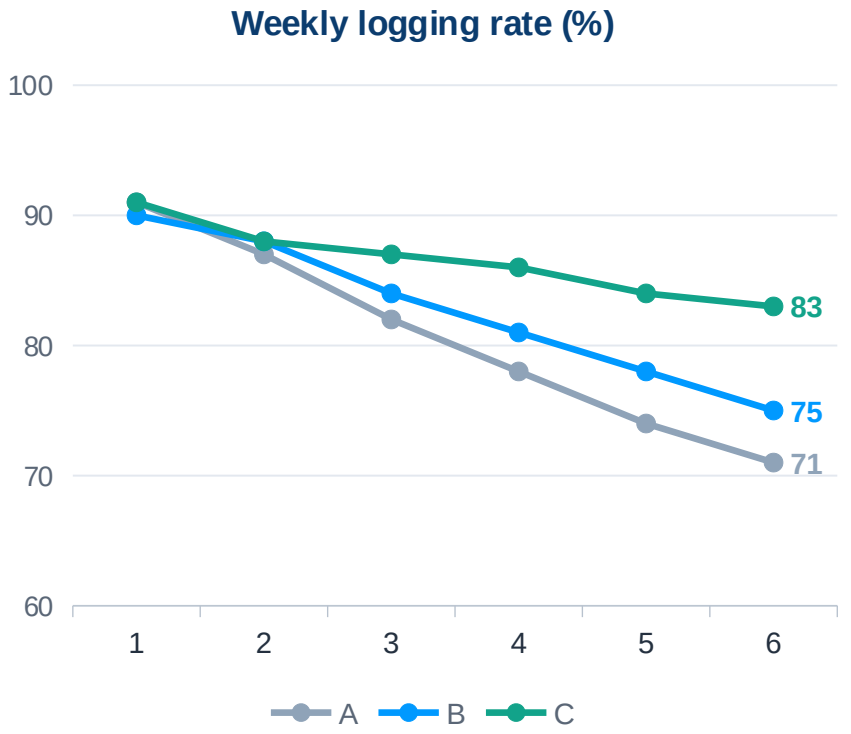
RQ2 · C vs B

Forecasts got better and stayed honest (RQ1)

- Fatigue error 0.84 → 0.62 after personalization
- 76% of outcomes fell inside the 80% interval

Results: well-being, joy, and logging





RQ3: small effect
C − B = +0.9 (n.s. after Holm); C − A = +1.7 (exploratory)

Joy was kept
Enjoyment stayed at 3.8–3.9 in all groups

People kept logging
Group C: 83% in week 6 (exploratory)

※ 想定値 (projected) — 要差し替え

Design implications

Make uncertainty a feature

Render forecast ranges as possible selves; let resolution grow with the user's own data

Rebalance, don't restrict

In domains without one right answer, keep what people love and move the rest

Same advice, better framing

How a future is shown can change follow-through even when the advice is identical

Limitations and next steps

Short and small

6 weeks, ~65 per arm; habits take longer (median 66 days; Lally et al., 2010)

Bundled features

Group C saw both avatars and dot counts; we cannot yet separate them

Self-report

Logs and WHO-5 are self-reported; physiological signals could complement them

Sample

Mostly university students and women in Japan

Next: longer follow-up, factorial study of avatar vs. dots, and other domains (social media, gaming)

Toward a sustainable fan economy with platform partners

Thank you!



OshiFit

Possible-selves avatars for uncertainty-aware behavior change

- Show the future honestly
- Rebalance, don't restrict
- Let people choose

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